Elliptic Squares: Exact Local Equations, Period Jets, and Arithmetic Budgets for Moment Certificates

Dr. Muralidhara Subbarao
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Abstract

We study the local-equation and interpolation interface in *Elliptic squares and Zilber–Pink for curves in A2*. The arithmetic theorem is treated as a source result, not as a consequence of numerical period computation. We develop exact common-parameter jet representations, finite moment tests with explicit truncation and conditioning costs, and arithmetic budgets for changes of frame. These distinguish a certificate of nonidentity from an enclosure of a small residual and from an exact relation at an algebraic point. A boundary-branch certificate remains conditional on the source's uniform multiplicity and geometric nonidentity inputs. We explain why improved analytic conditioning alone need not preserve all-place height estimates or the arithmetic properties of a G-function. The work consists of scoped source analysis and proved finite-dimensional refinements; it does not independently verify the full arithmetic proof or strengthen the stated Zilber–Pink theorem. Contribution assessment: C2/5, explicit local certificates and coordinate-cost lemmas sharpen the computational interface without a new arithmetic finiteness theorem.

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1 Scope and Source Architecture

The source considers a reduced irreducible closed algebraic curve $C \subset \mathcal{A}_2$ over $\overline{\mathbb{Q}}$ which is Hodge generic. Its elliptic-square assertion is finiteness of points $s \in C(\overline{\mathbb{Q}})$ for which the principally polarized abelian surface A_s is isogenous to E^2 with E non-CM. The isogeny need not preserve the principal polarization. Equivalently the full geometric rational endomorphism algebra is $M_2(\mathbb{Q})$. The elliptic curve, polarization and isogeny degree vary. The source imposes no boundary hypothesis. Its full Zilber–Pink conclusion additionally uses companion results for products with a CM factor and for the case where the full endomorphism algebra is an indefinite quaternion division algebra, together with special-point finiteness and classification. An embedding of a quaternion algebra is not the same hypothesis as equality with the full endomorphism algebra.

The catalogue lists thirteen TeX files. The mapped build also includes `references.bib`; adding the release README gives the parent's fifteen-text authenticated snapshot inventory. Snapshot agreement is provenance, not a proof audit. The machine-readable read ledger specifies each actually displayed line interval, its complement and pre/post hashes. A structure map does not claim literal reading of every mapped file.

Source component	Mathematical role and audit boundary
Introduction	Elliptic-square theorem, companion inputs, height/orbit route, conventions and dependence of fixed choices.
Interpolation	Conditional height theorem on every relevant boundary branch; small selected-place costs versus global height.
Monodromy	Rank-five quadratic system, full and joint monodromy, finite exceptional Hecke correspondences. These geometric inputs are imported.
Incidences	Two auxiliary levels: a degenerate four-space and a nondegenerate four-space, separately saturated plane lattices, bad-center disk exclusions. These integral-geometric inputs are imported.
Heights	Small actual endomorphisms, trait lattices, global coordinates, sharper near-center estimates. Integral lattice assertions are not numerical period certificates.
Local equations	Common good-reduction comparison, actual traces, simultaneous Frobenius conjugation, grouping and all-cutoff tails.
Nonidentities	Arbitrary algebraic branches, continuation, Frobenius obstruction, exceptional correspondences, height assembly.
Finiteness	Discriminant/orbit estimates, all polarizations, counting and classification. The global counting conclusion is imported.
Interpolation appendix	Absolute-height algebra, boundary multiplicity, graph construction, actual-ideal interpolation, product formula/descent.

Source component	Mathematical role and audit boundary
Main, preamble, Unicode file, figure, bibliography	Assembly and notation; the figure is an incidence schematic, not a separate proof.

The private originals used here are the finite-period, variation-of-Hodge- structure and corrected residual sections of `TR-item-011-astra.tex`, and the local moment/remainder and variable-coefficient jet arguments of Ram Calculus volume 1, revised chapter 3. Their surrounding Hodge problems are not hypotheses of the elliptic-square source. In particular, the report does not transfer a cycle-algebraization assertion to the present problem. Included chapter files not displayed in the read ledger remain unread. Private proof text was kept local; public searches used public titles only.

2 Exact Equations and Two Different Height Scales

Fix the source's fine-level cover, rational frame and actual center t_0. Write $x(t_0) = 0$, $a = x(t) \neq 0$, $d = \max(2, [K(t) : K])$ and $h = h(t) \geq 2$. Only the inverse branch through t_0 is used, not every zero of x. On the rank-five trace-zero Rosati-symmetric operator system the pairing is $\langle D, E\rangle = \mathrm{Tr}(DE)$. Let $\mathbf{B} = (B_1, B_2)$ be the center pair, $\mathbf{P} = (P_1, P_2)$ the target pair and $Y(a)$ horizontal back-transport, normalized by $Y(0) = I$. Matrix pairing means the four entries $\langle B_i, Y(a)P_j\rangle$.

The source produces, with accompanying base-coordinate equations, three types of *exact* equations:

$$\langle \mathbf{B}, Y(a)\mathbf{P}\rangle = \mathbf{r}, \qquad \mathbf{r} \in M_2(\mathbb{Q}), \tag{1}$$

$$\langle \mathbf{B}, Y(a)\mathbf{P}\rangle = \langle \mathbf{B}', Y'(aR)\mathbf{P}'\rangle, \qquad R = a'/a, \tag{2}$$

$$\langle \mathbf{B}, Y(a)\mathbf{P}\rangle = \langle I_0^{-1}\mathbf{B}', Y(a)\widehat{\mathbf{P}}\rangle, \qquad \widehat{\mathbf{P}} = I_t^{-1}\mathbf{P}'. \tag{3}$$

Here a prime denotes a conjugate system, not differentiation. In (2), Frobenius conjugates both factors of each trace. It disappears from the equation after taking traces. For the finite exceptional Hecke list, the transport square $I_0 Y(a) = Y'(a')I_t$ gives (3). The list of multipliers is fixed before choosing the auxiliary levels. Fixed denominators of $\widehat{\mathbf{P}}$ come from actual fixed-multiplier isogenies, not from rounding a computed matrix.

At retained good-reduction places the comparison is rational and allows ramification; it does not assert an integral crystalline comparison. The separately saturated lattices used in the auxiliary-level incidence are a different assertion, transported by prime-to-characteristic specialization. The chosen small endomorphism vectors need not be bases of these lattices modulo the auxiliary prime.

For a number field F use

$$h_{\mathrm{aff}}(z) = \sum_v w_v \log \max(1, |z_1|_v, \ldots, |z_n|_v), \qquad w_v = [F_v : \mathbb{Q}_v]/[F : \mathbb{Q}].$$

At a complex place use ordinary modulus and weight $2/[F : \mathbb{Q}]$; at a finite place $|p|_v = p^{-1}$. Selected embeddings carry the corresponding split weights. Pullback to an extension preserves their sums. The essential estimates, after selecting a group of embeddings V, are

$$h_{\mathrm{aff}}(a, U) \leq Cd^C h, \qquad \sum_{v\in V} w_v \log |a|_v^{-1} \geq h/(Cd^C), \tag{4}$$

$$\sum_{v\in V} w_v \log^+ \|U\|_v \leq Cd^C(1 + \log h)^C. \tag{5}$$

The distinction is decisive: an $O(h)$ global coordinate bound cannot replace the polylogarithmic selected-place cost. In particular $h(R) \le Ch$ globally, whereas $|R|_v = 1$ on the group in (2). There are polynomially many ordered conjugate tuples; the degree of a normal closure is not the number being counted.

Writing each scalar expression as $E(a, U) = \sum_{j\ge 0} a^j e_j(U)$, the source also requires degree and coefficient-height bounds and, for *every* $\ell \ge 0$,

$$\left|\sum_{j\ge \ell} a^j e_j(U)\right|_v \le |a|_v^{\ell}\, e^{(\ell+1)b_v}, \qquad \sum_{v\in V} w_v b_v \le C d^C (1+\log h)^C. \tag{6}$$

A successful computation at one cutoff does not verify this quantifier. At spread-out primes, the source uses $n!Y_n$ integral and $\beta_p = \log p/(p-1)$; retaining $-\log|a|_v > 2\beta_p$ gives the entire tail estimate by the ultrametric inequality. No bound on the number of supersingular proximity primes in terms of d alone is substituted.

3 Common-Parameter Jet Algebra

We formulate a local representation independently of the imported geometric construction. Let k be a characteristic-zero field and let $A(x) \in M_r(k[[x]])$. For the column convention $Z' = AZ$, $Z(0) = Z_0$, the unique formal solution satisfies

$$(n+1)Z_{n+1} = \sum_{i=0}^{n} A_i Z_{n-i}. \tag{7}$$

For a row convention the factors occur on the other side. Neither commutation of the coefficients nor a constant-coefficient exponential is assumed.

Proposition 3.1 (Exact common-parameter coefficients)**.** *For fixed bilinear forms and center vectors, all coefficients through degree N of (1)–(3) are obtained by (7) followed by finite matrix multiplication. In (2) the degree-n contribution from the second transport is $R^n\langle B_i', Y_n' P_j'\rangle$. With R, the operator slots and base slots treated as independent variables, each coefficient has polynomial degree at most $n + c$, where c is fixed. This assertion makes no claim that arbitrary values of those slots are endomorphisms.*

Proof. Coefficient comparison in $Z' = AZ$ gives the recurrence and uniqueness by induction. Substitution $x = aR$ gives $Y'(aR) = \sum_{n\ge 0} a^n R^n Y_n'$ in the formal power-series ring over $k[R]$. Contracting with a fixed center vector and a variable target vector adds one degree; the analogous base equations add no more. Fixed products, when present, add only their fixed number of slots. No field embedding or evaluation is involved in this identity. □

Proposition 3.2 (A conservative rational denominator budget)**.** *Assume $A_i \in D^{-1}M_r(\mathbb{Z})$ for $0 \le i < N$, with integer $D \ge 1$, and $Z_0 \in M_r(\mathbb{Z})$. Then $D^n n! Z_n$ is integral for $0 \le n \le N$. For a product of finitely many such series, the degree-n coefficient has the same denominator bound. At a prime not dividing D this recovers the factorial bound; a coordinate rescaling may enlarge the exceptional prime set.*

Proof. Multiplying the recurrence at index n by $D^{n+1}n!$, the summand with $A_i Z_{n-i}$ is integral by induction because $D^i n!/(n-i)!$ is an integer after the one denominator from A_i is cleared. Products use the integer multinomial factor $n!/(n_1!\cdots n_s!)$, and the powers of D sum to D^n. This proves a common denominator, not an optimal one. □

Remark 3.3. The hypothesis on all A_i up to the cutoff is explicit. A general rational connection does not automatically have a single denominator independent of its Taylor degree over $\mathbb{Z}$; one can use a cutoff-dependent D, or the source's integral local models at good primes. No exponential denominator property defining a G-function follows merely from the factorial estimate.

4 Moment Tests with Certified Tails

In this section only, fix a complex branch and let f be a scalar contraction of a transport or a scalar local equation with the other coordinates fixed. All norms are Euclidean or the induced operator norm. The following finite moments are an analytic diagnostic, not a replacement for equations valid at all the selected places.

Theorem 4.1 (Circular moment certificate)**.** *Suppose $f(z) = \sum_{n\geq 0} c_n z^n$ is analytic on a neighborhood of $|z| \leq R_0$ and $\sup_{|z|=R_0} |f|(z) \leq H$. Choose $0 < r < R_0$, an integer $q \geq 1$, and $\zeta = e^{2\pi i/q}$. For $0 \leq k < q$ define*

$$d_k = \frac{r^{-k}}{q} \sum_{j=0}^{q-1} f(r\zeta^j)\zeta^{-jk}.$$

Then

$$d_k = c_k + \sum_{\ell\geq 1} c_{k+\ell q} r^{\ell q}, \qquad |d_k - c_k| \leq H R_0^{-k} \frac{(r/R_0)^q}{1-(r/R_0)^q}. \tag{8}$$

If each sample is known with complex absolute error at most η, and $\widehat{d}_k$ is evaluated with additional absolute error ϵ_k, then

$$\left|\widehat{d}_k - c_k\right| \leq H R_0^{-k} \frac{(r/R_0)^q}{1-(r/R_0)^q} + r^{-k}\eta + \epsilon_k =: \Delta_k. \tag{9}$$

In particular $\left|\widehat{d}_k\right| > \Delta_k$ proves $f \not\equiv 0$.

Proof. Uniform convergence on $|z| = r$ permits substitution into the finite sum. The average of $\zeta^{j(n-k)}$ is one when $n \equiv k \pmod q$ and zero otherwise. Since $0 \leq k < q$, the surviving indices are $k + \ell q$, $\ell \geq 0$. Cauchy's coefficient bound $|c_n| \leq H R_0^{-n}$ gives the geometric tail. The sum of the absolute weights of the sampling functional is r^{-k}, giving the sample-error term. Finally a coefficient whose certified disk excludes zero is nonzero. □

This is an exact finite moment operator on polynomials of degree less than q, with an explicit analytic remainder on larger spaces. It is a direct specialization of the private chapter's moment-plus-remainder principle. For the scaled coefficients $r^k c_k$ the sample error is η for every k; for unscaled high jets it grows like r^{-k}. Smaller circles reduce aliasing but amplify sample error. Thus shrinking a panel is not unconditionally beneficial.

For a matrix transport $Y' = AY$ with $Y(0) = I$ and $\sup_{|z|\leq R_0} \|A(z)\| \leq M$, radial integration gives $\|Y(z)\| \leq e^{MR_0}$ on the disk. Consequently a contraction $b^*Y(z)p$ has $H \leq \|b\|\,\|p\|\,e^{MR_0}$. For the paired expression one must also have $|R|\,R_0$ inside the convergence disk of the primed transport and bound its contribution separately. The selected nonarchimedean assertion $|R|_v = 1$ gives no such complex-disk radius automatically.

Proposition 4.2 (Certified moment reconstruction)**.** *Let $\lambda_1, \ldots, \lambda_m$ be bounded linear functionals on a normed analytic function space, and let $f = p + \rho$ with $p = \sum_{k=0}^{N} c_k z^k$, $\|\rho\| \leq \tau$. Write $V_{ik} = \lambda_i(z^k)$*

and suppose the combined functional map has norm at most L. Assume a computed matrix $\widehat{V}$ has $\left\|V-\widehat{V}\right\| \le e_V$, with a certified $s \le \sigma_{\min}(\widehat{V})$ and $s > e_V$. If the data satisfy $\left\|\widehat{b}-(\lambda_i(f))_i\right\| \le e_b$, then any coefficient vector $\widehat{c}$ satisfies

$$\|c-\widehat{c}\| \le \frac{\left\|\widehat{b}-\widehat{V}\widehat{c}\right\| + e_b + L\tau + e_V\,\|\widehat{c}\|}{s-e_V}. \tag{10}$$

The matrix has $N+1$ columns, and the least singular value here means the lower bound on all of $\mathbb{C}^{N+1}$.

Proof. Insert $\widehat{b}$ and $\widehat{V}\widehat{c}$ into $V(c-\widehat{c})$. The four remaining terms have the four numerator bounds. The singular-value perturbation inequality gives $\|Vx\| \ge (s-e_V)\,\|x\|$, proving the result. □

Polynomial exactness alone gives no estimate for ρ. Likewise, differentiating an integral moment formula requires derivatives of its variable coefficients and of its remainder. For example $(I-bD^2)(I+bD^2)f = f - bb''f'' - 2bb'f''' - b^2f''''$. The last three terms cannot be discarded by declaring b locally constant. No such frozen inverse is used in (10).

5 Boundary Nonidentity and a Smaller Search

The source's interpolation condition quantifies over every irreducible closed algebraic W containing the target, with a nonconstant, and every prime divisor on its finite normalization above a component of $W \cap \{a=0\}$. At least one equation must be a nonzero analytic germ on the branch at a general point of that divisor. It is insufficient to check the original curve, a single specialization of the operator slots, or the restriction obtained by merely setting $a=0$.

The source includes the base equations $u_\nu - f_\nu(a) = 0$ and, when appropriate, $u'_\nu - f'_\nu(aR) = 0$. A putative identity then gives actual algebraic curve projections. Continuation around loops varies the horizontal transports while returning algebraic coordinates. Full monodromy excludes a constant nonzero matrix coefficient; product monodromy deals with the paired case outside the fixed Hecke list. For the exceptional case it forces $B_i \otimes P_j = (I_0^{-1}B_i') \otimes \widehat{P}_j$ and a common scalar. The rational nonzero center trace norms make that scalar ± 1. At the actual target, faithfulness of realization turns these into equalities of actual rational endomorphisms.

The source excludes $+1$ because a rational scalar would have square p/m_* with odd p-adic valuation. For -1, an integral orthogonal splitting at the odd auxiliary prime makes the rank-four minus eigenspace unimodular, contradicting the imposed degenerate reduction of the two separately saturated planes. None of these contradictions is established by a numerical singular value of a period matrix.

Theorem 5.1 (One-parameter avoidance of boundary cancellations)**.** *Let $E_1,\dots,E_s$ be analytic germs along $B \ge 1$ specified prime divisors $D_1,\dots,D_B$ of a normal characteristic-zero variety. Assume their finite orders and leading coefficients are defined over the respective residue function fields, and on each divisor at least one E_α is not identically zero. Put $\nu_b = \min_\alpha \operatorname{ord}_{D_b} E_\alpha$, ignoring identically zero germs. There is an integer*

$$0 \le t \le B(s-1) \quad \textit{such that} \quad E_t = \sum_{\alpha=1}^{s} t^{\alpha-1}E_\alpha \quad \textit{satisfies} \quad \operatorname{ord}_{D_b} E_t = \nu_b \quad (1 \le b \le B).$$

The first coefficient is defined to be 1, also when $t=0$. For $s=1$ take $t=0$ and $E_t = E_1$. For $s \ge 2$ the coefficients are nonnegative integers bounded by $\max(1, B(s-1))^{s-1}$.

Proof. Fix a local uniformizer at the generic point of D_b. The coefficient of its ν_b power in E_t is the nonzero polynomial $p_b(t) = \sum_\alpha \ell_{b,\alpha} t^{\alpha-1}$ over the residue function field, where $\ell_{b,\alpha}$ is the leading coefficient if the order equals ν_b and zero otherwise. It has degree at most $s-1$, hence at most $s-1$ roots in that field. Characteristic zero makes the $B(s-1)+1$ candidate integers distinct. The union of the forbidden roots has at most $B(s-1)$ elements. At a remaining integer no leading order cancels on any divisor. The coefficient bound follows immediately. □

Corollary 5.2 (Application to the source multiplicity step)**.** *Assume the source's ordinary multiplicity input gives $\nu_b < M \operatorname{ord}_{D_b} a$ simultaneously, and the number of normalized boundary divisors is at most $\deg W$. The combination in Theorem 5.1, truncated before a^M, has a pole after division by a^M on every such divisor. For fixed number s of equations, the search uses at most $(\deg W)(s-1)+1$ candidates. Its coefficient height is $O_s(1+\log \deg W)$. It preserves the source's polynomial complexity class.*

Proof. The analytic remainder after the truncation is divisible by a^M, since the coefficient functions are regular there. It has order at least $M \operatorname{ord} a$, strictly larger than the combination's order. The truncation therefore has the same first order; division produces a negative order. The candidate count and height follow from the theorem. □

The M040 source uses the full grid $\{0, \dots, \deg W\}^s$, whose existence argument is already correct. The companion source [2], interpolation lines 653–663, already uses this one-parameter integer-root-avoidance device. The parent identified that attribution; we then read the original lines 639–672 independently. Our formulation transfers the device to M040 and spells out its finite candidate and coefficient budgets. This adaptation reduces the number of candidate combinations but allows larger individual coefficients. It is useful only when leading-coefficient tests are available exactly; it is not an algorithm that discovers all boundary branches for free. Nor is this a claim of historical novelty for polynomial root avoidance. At finite places the nonnegative integer coefficients have norm at most one. At infinity the all-cutoff tail cost increases by at most

$$\log \left(\sum_{\alpha=1}^{s} t^{\alpha-1} \right) \le \log s + (s-1) \log \max(1, B(s-1)). \tag{11}$$

The total weighted increase is no larger than this because the total archimedean weight is one. Thus the refinement respects the selected-place budget, not just the existence of a complex nonzero germ.

5.1 Why exact finite jets need a multiplicity premise

On a transverse normalized curve write $a = u^e v(u)$ with $e \ge 1$ and $v(0) \ne 0$. If a known theorem says that every nonzero germ in a declared class has order at most K, then its coefficients through u^K all vanishing is equivalent to identity within that class. This follows by contraposition. Without that theorem u^{K+1} is a counterexample. Without exact coefficient tests, a small constant is another counterexample.

The appendix supplies its uniform estimate by representing an expression as $r\,y$ in a fixed-rank connection on a varying algebraic curve. The rows $r_{j+1} = \delta r_j + r_j B$ stabilize in their function-field span. The saturated kernel over $\mathbb{C}\{u\}$ is a direct summand preserved by the ordinary connection. A nonzero horizontal quotient vector is primitive; Smith exponents of the derivative-row matrix then bound the first order. The ratio $\operatorname{ord} f / \operatorname{ord} a$ is unchanged by finite ramified pullback. These are the reasons a uniform multiplicity bound is plausible in the source's precise setting. Our local finite tests do not independently establish its bounds for every varying W.

5.2 Actual ideals, not residual ideals

The graph of $q = a^{-M}E_{<M}$ has projected boundary of dimension at most $\dim W - 2$. This is a statement about its projection; the whole graph boundary may have larger fibers. Imposing $q + e_M(U) = 0$ fixes the extra coordinate and gives the needed dimension estimate. The source constructs an auxiliary polynomial

$$P \in (I, a^n, F_n), \qquad F_n = q + \sum_{j=0}^{n-1} a^j e_{M+j}(U), \tag{12}$$

with I the actual generated ideal, possibly nonradical. Membership in its radical would not preserve the a^n factor. The exact identity $E(a, U) = 0$ makes F_n an exact higher tail. With only $|E(a, U)| \le \varepsilon_v$ it instead contributes an additional term $|a|_v^{-M}\varepsilon_v$. This can overwhelm $|a|_v^n$ for arbitrarily large n. No fixed nonzero numerical tolerance can support the all-cutoff product-formula descent.

At selected places the certificate degrees multiply the small local coordinate costs. Elsewhere one evaluates P itself, of degree L, not the potentially larger certificate. Choosing $n = cL$ after fixing the global-height coefficient gives $0 \le (-n/A(d) + LC(d))h + D(d)(1 + \log h)^B$ if $P(z') \ne 0$. Otherwise exact vanishing permits a proper algebraic cut and dimension descent. A small residual does not choose either branch of this dichotomy.

6 Frame Changes: Analytic and Arithmetic Costs

6.1 Differential and pairing conventions

Let $y' = Ay$ on a simply connected disk and let G be analytic and invertible there. Under $y = Gz$,

$$z' = \widetilde{A}z, \qquad \widetilde{A} = G^{-1}AG - G^{-1}G'. \tag{13}$$

The prime on G here is differentiation. If Q is the bilinear pairing matrix, its transformed matrix is G^TQG, not necessarily Q. A transported dual functional changes as well. For a Hermitian error metric use G^*MG, not the bilinear polarization matrix. The source's quadratic trace form on the whole complex representation is not a positive error norm.

Proposition 6.1 (Jet changes are block triangular)**.** *Let $G(x) = \sum_{j\ge0} G_j x^j$ with G_0 invertible. For Taylor-coefficient jets through order N, multiplication by G is the block lower triangular matrix*

$$\mathcal{G}_N = (G_{i-j})_{0\le j\le i\le N}, \qquad \det \mathcal{G}_N = (\det G_0)^{N+1}.$$

In the block maximum norm,

$$\|\mathcal{G}_N\| \le \sum_{j=0}^{N} \|G_j\|, \qquad \left\|\mathcal{G}_N^{-1}\right\| \le \sum_{j=0}^{N} \|H_j\|, \quad G^{-1}(x) = \sum H_j x^j.$$

Thus finite-jet rank is preserved, while conditioning and denominators can change.

Proof. The coefficient of x^i in Gz is $\sum_{j=0}^{i} G_{i-j}z_j$. This is the stated triangular matrix; its diagonal consists of $N + 1$ copies of G_0. Truncation commutes with formal multiplication, so the inverse is the analogous matrix for G^{-1}. The row-sum bounds give the norm estimates. □

This concerns coefficient jets, not unscaled derivatives: derivative jets introduce the usual factorial conjugations. In an analytic ODE, frame invertibility is required on the whole panel. A frame regular only away from $a = 0$ can create a boundary pole and change the order bookkeeping. Panel continuation must carry its branch and local-system basis; closeness of two computed values does not identify their monodromy.

6.2 All-place bounds for an algebraic frame

Theorem 6.2 (Constant algebraic coordinate budget)**.** *Let $T \in \mathrm{GL}_n(F)$, $z \in F^n$, with the absolute weights of Section 2. Write $H_T = h_{\mathrm{aff}}((T_{ij})_{i,j})$. Then*

$$h_{\mathrm{aff}}(Tz) \le h_{\mathrm{aff}}(z) + H_T + \log n, \tag{14}$$

$$h_{\mathrm{aff}}(T^{-1}) \le nH_T + \log(n!), \tag{15}$$

$$h_{\mathrm{aff}}(z) \le h_{\mathrm{aff}}(Tz) + nH_T + \log(n!) + \log n. \tag{16}$$

For any weighted selected set V, put $K_V(T) = \sum_{v\in V} w_v \log \max(1, \max_{ij} |T_{ij}|_v)$. Then

$$\sum_{v\in V} w_v \log \max(1, \|Tz\|_{v,\infty}) \le \sum_{v\in V} w_v \log \max(1, \|z\|_{v,\infty}) + K_V(T) + \log n. \tag{17}$$

Here $\|\cdot\|_{v,\infty}$ is the maximum coordinate absolute value, including at the archimedean places.

Proof. At an archimedean place, a row sum costs at most n; at a finite place the ultrametric inequality costs no factor. Taking positive logarithms and summing proves the forward and selected-place estimates. For the inverse use the projective tuple $[\det T : (\operatorname{adj} T)_{ij}]$, whose height is $h_{\mathrm{aff}}(T^{-1})$. Each coordinate is bounded locally by $\max(1, \max |T_{ij}|_v)^n$, with an extra factor at most $n!$ at infinity. Projective scaling by $\det T$ costs zero in the total sum by the product formula. This proves (15), and applying (14) to T^{-1} proves the last bound. □

Corollary 6.3 (An admissible target-dependent preconditioner)**.** *In the fixed ambient dimension of the source, replacing a coordinate block U by TU preserves the numerical height scales if the additional global and selected-place costs of T and the inverse required by the rewritten equations have the corresponding bounds $Cd^C h$ and $Cd^C(1 + \log h)^C$. Rewriting the equations must also preserve their coefficient heights, fixed-germ form and degree bounds. A fixed algebraic matrix has bounded costs and meets the coordinate part of these requirements.*

Proof. Apply Theorem 6.2 to both directions. Substituting the inverse matrix into a polynomial of degree $j + c$ multiplies its height cost by at most that degree, with a polynomial combinatorial term in fixed dimension. This preserves the permitted polynomial coefficient-height form when the inverse coefficient costs have that form. The separate fixed-germ and geometric conditions are not consequences of the coordinate inequalities and remain explicit premises. □

A target-dependent change of equation coefficients must in particular have polylogarithmic-in-h coefficient cost, not merely $O(h)$, to preserve the source's coefficient hypothesis. One cannot use the global coordinate estimate alone to justify such a change. For $T = 2^{-k}I$, the real operator norm tends to zero, but the cost at the prime 2 is $k \log 2$; indeed $H_T = k \log 2$. This example isolates the difference between real rescaling and arithmetic improvement.

For a rational matrix $T = A/L$ with A integral and $L \ge 1$, $T(D^{-1}\mathbb{Z}^n) \subset (LD)^{-1}\mathbb{Z}^n$. Conversely $T^{-1} = L \operatorname{adj}(A)/\det A$; thus inverse denominators divide a bound involving $|\det A|$, not just L. With $|A_{ij}| \le H_A$, the elementary determinant estimate is $|\det A| \le n! H_A^n$. These are explicit conservative denominator costs. A floating-point orthogonalization has no such rational denominator certificate by default.

6.3 Analytic improvement without an arithmetic claim

For any differentiable approximation p to $y' = Ay$ on $[0,T]$, put $r = p' - Ap$. If $\mu(t) = \lambda_{\max}((A(t) + A(t)^*)/2)$, then

$$\|y(t) - p(t)\| \le e^{\int_0^t \mu} \|y(0) - p(0)\| + \int_0^t e^{\int_s^t \mu} \|r(s)\| \, ds. \tag{18}$$

For a positive differentiable Hermitian metric $M(t)$ the analogous formula uses the corresponding endpoint norms and

$$\mu_M = \tfrac{1}{2}\lambda_{\max}(M^{-1/2}(M' + A^*M + MA)M^{-1/2}).$$

To prove these statements, differentiate the squared norm of a homogeneous solution to bound its propagator, then use variation of constants on the defect equation. The term M' is essential. This reproduces the corrected private-proof route, not a new ODE theorem.

For example, a skew-Hermitian A gives $\mu = 0$, so the defect error is bounded by the initial error plus $\int \|r\|$, without the exponential of $\int \|A\|$. This can improve certified complex sample errors in Theorem 4.1. It does not supply a finite-place tail.

Even an everywhere invertible analytic gauge need not preserve G-function arithmetic. Start with the constant solution $y = 1$ and take $G(x) = e^x$, so $z = e^{-x}$ satisfies the rational ordinary equation $z' = -z$. Its first $N+1$ Taylor coefficients have least common denominator $N!$, which exceeds C^N for every fixed C for large N. It therefore fails the exponential common-denominator condition for a G-function. Rational differential equations and analytic invertibility alone are insufficient. An arbitrary transcendental constant gauge may even destroy algebraicity of the initial coefficients.

7 What a Small Residual Can Certify

There are three different conclusions: nonzero germ, exact algebraic vector equality, and exact zero of the entire local equation at the specified arithmetic target. The first can follow from a coefficient enclosure excluding zero. The second needs arithmetic separation. The third is supplied in the source by actual endomorphisms and comparison, not by either test alone.

Proposition 7.1 (Full-space lattice separation with errors)**.** *Let $v \in D^{-1}\mathbb{Z}^n$, with proved integer $D \ge 1$. Let $T : \mathbb{R}^n \to \mathbb{C}^m$ be real-linear with $\|Tx\| \ge s\|x\|$ for every $x \in \mathbb{R}^n$, where $s > 0$. Suppose $\left\|T - \widehat{T}\right\| \le \eta$, $\|v\| \le B$, and a computed data vector $\widehat{r}$ satisfies $\left\|\widehat{r} - \widehat{T}v\right\| \le \epsilon$. If*

$$\|\widehat{r}\| + \epsilon + \eta B < s/D, \tag{19}$$

then $v = 0$.

Proof. The left side bounds $\|Tv\|$. A nonzero vector in the declared lattice has Euclidean norm at least $1/D$, which would imply $\|Tv\| \ge s/D$, a contradiction. □

This is the private lattice-gap argument with the operator and data errors made explicit. It applies only to a declared full rational comparison space. A residual of the form $\langle B, Y(a)P\rangle - r$ is not known to lie in $D^{-1}\mathbb{Z}$ merely because the formal coefficients of Y are algebraic. No application of Proposition 7.1 to arbitrary period values is asserted.

The strict inequality is sharp as a uniform criterion: for $T = I$, $v = (1/D, 0, \ldots, 0)$ has norm exactly $1/D$ and is nonzero. Without a denominator bound the vectors $(1/j, 0, \ldots, 0)$ approach zero; without full-space injectivity a nonzero unmeasured vector has zero residual. For algebraic

number-field coordinates one needs a specified lattice in all embeddings, with its covolume or shortest-vector bound. One complex embedding alone has no rational lattice gap of the form asserted above.

8 Literature, Comparison, and Limitations

The public literature check on October 8, 2026 was targeted, not an exhaustive priority search. The source release is [1]. We independently inspected the following primary items at the stated scope. Papas [3], Theorem 1.5 and Corollary 1.6, assumes an everywhere potentially good split center: its height bound retains the supersingular proximity count, and its finiteness statement bounds that count. The removal of the potentially-good assumption in Theorem 1.8 is conditional on two comparison conjectures. It should not be described as the unrestricted elliptic-square theorem considered here.

Daw–Orr–Papas [4], version 2 of February 25, 2026, Theorem 1.3 treats curves in $Y(1)^3$ whose closure meets a modular point in the boundary. Their Theorem 1.6 instead restricts the supersingular proximity exponent of modular points relative to a fixed modular center. These are not all-curve statements for $\mathcal{A}_2$. Papas [5] concerns quaternionic-multiplication relations in one-parameter families of principally polarized abelian surfaces; only its abstract and version metadata were inspected here. No theorem from that paper is used as an unverified substitute.

The derivative-span multiplicity method is classical. Bertrand–Beukers [6], Theorem 1 on printed pages 181–182, starts with a rational linear differential system and a polynomial expression in a solution analytic at zero. It bounds the order of a nonzero expression using the dimension of its derivative span and the polynomial degrees. Our one-parameter coefficient search is not a new multiplicity theorem. The source's appendix additionally handles varying algebraic curves and normalized boundary branches with its own ordinary-point argument. We read that selected chain, while the external effective algebra estimates and its full global arithmetic deployment remain imported.

Method	Advantage	Cost or limitation
Exact recurrence	Common-parameter coefficients and reproducible denominators	Large coefficients; a finite zero jet needs a proved multiplicity bound to imply identity.
Circular moments	Polynomial exactness and explicit aliasing error; scaled jets control sample noise	Complex branch and radius must be certified; high unscaled jets amplify noise; no automatic arithmetic values.
General moment matrix	Works with application-specific bounded functionals	Tail and lower frame bound are required on the declared space, with every error counted.
One-parameter combination	At most $B(s-1)+1$ candidates instead of a full grid	Larger coefficients; exact boundary leading terms still needed.
Rational preconditioning	Preserves an exact algebraic description with recordable heights	Bad primes, inverse denominators and selected-place costs can erase an analytic gain.
Analytic metric or gauge	May sharply improve complex defect estimates	Does not by itself preserve the source's fixed arithmetic frames or G-function coefficient bounds.

Method	Advantage	Cost or limitation
Lattice-gap equality	Turns a rigorous small residual into exact zero under explicit separation	Rationality, denominator and full-space injectivity must already be proved.

8.1 Feasible extensions and work still required

An implementation can keep two frames: a fixed arithmetic frame for all-place arguments and an analytic frame for validated integration, with exact or enclosed transition matrices and full conversion errors. For a fixed numerical instance this may improve precision requirements without rewriting the arithmetic theorem. Optimizing rational preconditioners against both conditioning and $K_V(T)$ is a sensible finite problem once the actual data and places are known.

The one-parameter combination theorem remains valid over any characteristic-zero residue fields and for any finite list of branch orders; it does not need the five-dimensional orthogonal representation. Extending the source height theorem to new geometries would still require uniform multiplicity, actual local relations, small selected-place coordinates, and nonidentity on every descendant variety. None is supplied merely by this field-independent combinatorial lemma.

Future computation should report a coefficient certificate, a denominator certificate and a branch certificate separately. It should benchmark the candidate-count saving against increased coefficient sizes, not just count arithmetic operations. A genuine arithmetic improvement would have to change a height exponent, a geometric hypothesis, or an orbit bound under fully verified new input. No such improvement is claimed here.

9 Evidence and Contribution Assessment

The accompanying evidence records the literal source/private intervals, unread complements, file hashes, public-search scope, finite checks, compiler outcomes and every final page inspected. Tests concern finite algebraic models and numerical sanity checks of proved bounds. They do not certify monodromy, integral models, all normalized branches, or all algebraic targets. Parent checks and source snapshot authentication are separately attributed and are not counted as author-run mathematical tests.

The contributions supported here are Proposition 3.1 and Proposition 3.2 (exact representations and a conservative denominator budget), Theorem 4.1 and Proposition 4.2 (explicit finite moment certificates), Theorem 5.1 and Corollary 5.2 (a smaller candidate family with all-place coefficient cost), and Theorem 6.2 with Proposition 7.1 (coordinate and exactness budgets). Their proofs use elementary algebra and analysis, with the private originals credited for the moment, residual and lattice-gap routes. The conditional branch application explicitly imports the source's multiplicity and geometric nonidentity inputs.

The assessment is C2/5: useful, source-specific, proved local refinements and an auditable computational interface. Applicability L3 is the catalogue's separate measure of relevance. Neither label establishes novelty, a full-proof audit, or a breakthrough in Zilber–Pink.

References

[1] OpenAI. *Elliptic squares and Zilber–Pink for curves in A2.* September 24, 2026 release, sections 2, 5–7 and interpolation appendix at the line scopes in the read ledger. Public release directory.

The report uses the pinned local snapshot, not an assumption about a live branch.

[2] OpenAI. *Quaternionic division points on curves in the Siegel threefold*. September 24, 2026 release. Only interpolation lines 639–672 were read for the root-avoidance attribution here; no companion audit is claimed. Public companion release directory.

[3] G. Papas. *On the v-adic values of G-functions I: Splittings in* $\mathcal{A}_2$. arXiv:2510.11806v1 (2025). https://arxiv.org/html/2510.11806v1.

[4] C. Daw, M. Orr and G. Papas. *Some new cases of Zilber–Pink in* $Y(1)^3$. arXiv:2510.09603v2, February 25, 2026. https://arxiv.org/html/2510.09603v2.

[5] G. Papas. *On the v-adic values of G-functions III*. arXiv:2510.19683v1 (2025). Abstract-level comparison only. https://arxiv.org/abs/2510.19683.

[6] D. Bertrand and F. Beukers. *Équations différentielles linéaires et majorations de multiplicités*. Ann. Sci. École Norm. Sup. (4) 18 (1985), 181–192. Theorem 1 and its introductory hypotheses. https://www.numdam.org/item/10.24033/asens.1488.pdf.

[7] M. Subbarao. Private original `TR-item-011-astra.tex`, finite-period certificates, variation-of-Hodge-structure jets, corrected Taylor/defect and lattice-gap sections. Local access only; exact displayed intervals and SHA-256 in the read ledger.

[8] M. Subbarao. *Ram Calculus*, volume 1, revised chapter 3, `ch03-local-moments-rev2.tex`. Local differential representation and remainder, differentiated moment jets and frozen-inverse defects. Local access only; included chapter files are not implicitly counted as read.